

Wide-Scale Sampling with Hard-Argmin Selection and CBF Projection for Safe Quadrotor Navigation in Narrow Passages

immediate

Abstract

Safe quadrotor navigation through narrow passages is a demanding real-time control problem: nonlinear dynamics couple with non-convex obstacle geometry under hard collision constraints and bounded compute. Sampling-based NMPC (MPPI, CEM) is attractive for its derivative-free obstacle handling, but the conventional single-source pipeline fails on tight passages, where the nominal control points into an obstacle and every candidate inherits the bias. We propose TSH-NMPC, combining three components: **(P1)** wide-scale Gaussian sampling ($\sigma \geq 5$ N per axis) to escape the biased nominal; **(P2)** hard-argmin selection (no soft weighting) to preserve escape-candidate quality; **(P3)** a DT-CCBF safety projection to filter unsafe wide-scale candidates. The three are individually empirically necessary and jointly effective within the narrow-passage regime studied here. On a paired Monte-Carlo benchmark ($N_{MC} = 80$) the controller at $K = 10$ achieves 96–99% success (Tables 2, 6) versus 66–68% for narrow-scale PD-Gaussian NMPC (McNemar $p < 0.001$), with comparable behaviour on two additional task families and three seeds. Under +50% mass mismatch, the wide-scale controller retains 77.5% success (single-seed, $N_{MC} = 40$), rising to 82.5% across three seeds ($N_{MC} = 120$), while PD collapses to 0%. Against CasADi+IPOPT collocation NMPC at $H = 20$, TSH-NMPC achieves success-rate parity ($p = 1.0$) at ≈ 6 ms per-call controller cost on a single CPU without iterative nonlinear programming and with approximately deterministic compute. A negative ablation at $N_{MC} = 300$ finds a 27 935-parameter learned prior significantly inferior to a random wide source ($p = 7.4 \times 10^{-6}$), identifying sampling *diversity* as the driving mechanism. The controller is simple to implement, has only two main hyperparameters (K and σ), and is suitable for embedded platforms without GPU acceleration.

Keywords: Nonlinear model predictive control; sampling-based MPC; control barrier functions; quadrotor obstacle avoidance; ablation study.

1. Introduction

Sampling-based NMPC — MPPI [13], CEM [2] — is a standard tool for online obstacle avoidance: rollout-and-score evaluation needs no convexity or differentiability, and per-step compute is bounded (K rollouts of horizon S) and trivially parallel.

The standard *single-source* formulation draws K first-step candidates from one Gaussian centered at a nominal. On tight passages the nominal often points *into* an obstacle and every candidate inherits the bias; increasing K yields diminishing returns. Single-source PD-Gaussian rollout NMPC achieves only 73.8% success at $K = 20$ on our narrow-passage benchmark (Section 5.2).

We identify a *three-pillar* mechanism for test-time scaling in non-convex obstacle environments: **(P1)** wide-scale perturbation ($\sigma \geq 5$ N) to escape the biased nominal; **(P2)** hard-argmin selection (no soft weighting) to preserve escape-candidate quality; **(P3)** DT-CCBF safety projection to filter unsafe wide-scale candidates. P1 and P3 are individually empirically necessary

for success and safety; P2 for competitive tracking accuracy; the three are jointly effective within the narrow-passage regime (Section 5.4). TSH-NMPC (Test-time Scaling with Hard-argmin) draws K PD-Gaussian candidates at $\sigma = 5$, scores by short rollout, hard-argmin selects, and CBF-projects. The sampler admits two equivalent forms: $R_{0,\sigma=5}$ (single-source-wide) and $R_{1,\sigma=5}$ (two-source 50/50 mixture of $\sigma_{\text{pd}} = 2$ and $\sigma_r = 5$); McNemar confirms equivalence ($p = 0.51$, Section 5.5.2): the lever is the noise scale, not the source partition.

1.1 Contributions

1. **A three-pillar mechanism for test-time scaling in non-convex obstacle environments.** Wide-scale sampling ($\sigma \geq 5$), hard-argmin selection, and DT-CCBF safety projection are jointly effective: removing P1 caps success at $\leq 68\%$ (Table 2); replacing P2 with soft-weighting (MPPI) degrades TErr by $4.4\times$; removing P3 yields 40/40 collisions. Under +50% mass the three-pillar controller retains $\geq 77.5\%$ success (PD: 0/40, Section 5.5).
2. **A rollout-based selector with monotone K -improvement** (Proposition 2) at 5.7 ms per-call cost at $K = 10$ on commodity CPU (Section 5.10).
3. **A diversity-over-accuracy ablation.** At $N_{\text{MC}} = 300$, a 27,935-parameter learned predictor is significantly *outperformed* by a random wide source (McNemar $p = 7.4 \times 10^{-6}$, Section 5.6): candidate *diversity* beats candidate *accuracy* when escape geometry is approximately isotropic.
4. **Success-rate parity with CasADi+IPOPT at deterministic compute budget, without iterative NLP.** Paired McNemar $p = 1.0$ on NARROW $N_{\text{MC}} = 80$ at $K \geq 20$; CasADi retains a $5\text{--}11\times$ TErr advantage and lower mean latency but data-dependent per-step cost (max = 35 ms) vs. TSH-NMPC’s $\mathcal{O}(K \cdot S)$ algorithmically data-independent compute (CV < 20%) (Section 5.7).

1.2 Paper Organization

Sec. 2: dynamics; 3: method; 4: theory; 5: experiments; 6: discussion; 7: conclusion.

1.3 Related Work

Sampling NMPC. MPPI [13, 14] uses importance-weighted averaging — a strength on race tracks but a liability in tight passages [4, 9]. CEM [2, 11] iteratively refits the top- β quantile. TSH-NMPC is one-shot: diversity is paid once in a fixed-budget pool rather than per iteration.

CBFs. Continuous-time CBFs [1] enforce forward-invariance; discrete-time variants [12] extend the guarantee to sampled-data execution. We use the off-the-shelf DT-CCBF as safety filter.

Learned warm-starts / diversity. Prior works replace the Gaussian by a learned proposal [7, 8, 10, 21] but rarely compare against isotropic-random at matched budget (Section 5.6). The narrow-passage difficulty [3, 6] is typically attacked at the planner level; our controller-level analogue shifts diversity to the inner loop. Analogues in MCTS [16], progressive widening [17], evolution strategies [18, 19], and best-arm identification [20].

Scenario optimisation. Randomized optimisation [24, 25] bounds violation probability as K grows. TSH-NMPC’s hard-argmin over K rollout-evaluated candidates is an instance — *without* the convexity assumption — and Proposition 2 gives the corresponding monotonicity guarantee.

2. Problem Formulation and Background

2.1 Quadrotor Abstract Translational Dynamics

The outer-loop translational guidance operates on a 6D state $x_t = [p_t^\top, v_t^\top]^\top \in \mathbb{R}^6$. The virtual control $u_t \in \mathbb{R}^3$ is the gravity-precompensated net force command; the inner-loop attitude controller is assumed fast enough ($T_{\text{att}} \ll \Delta t$) that flatness-mapped tracking does not affect the translational closed loop. Discrete-time dynamics:

$$x_{t+1} = A_d x_t + B_d u_t, \quad A_d = \begin{pmatrix} I_3 & \Delta t I_3 \\ 0 & I_3 \end{pmatrix}, \quad B_d = \frac{\Delta t}{m} \begin{pmatrix} 0 \\ I_3 \end{pmatrix}, \quad (1)$$

with body drag omitted in the nominal model (added in simulation, Section 5.1). Input saturation: $u_t \in [u_{\min}, u_{\max}]^3$, $u_{\max} = 15 \text{ N}$.

2.2 Discrete-Time Composite CBF (DT-CCBF)

Let $\mathcal{C}_i = \{x \in \mathbb{R}^6 : h_i(x) \geq 0\}$ be the obstacle-free set for obstacle i , with $h_i(x) = \|p - p_i\|^2 - r_i^2$. The unsafe set $\cup_i \bar{\mathcal{C}}_i$ is non-convex. We replace the per-obstacle minimum by the LSE composite

$$h(x) = -\frac{1}{\kappa} \log \sum_{i=1}^M \exp(-\kappa h_i(x)), \quad \kappa = 5, \quad (2)$$

with $h(x) \leq \min_i h_i(x)$ and $h(x) \rightarrow \min_i h_i(x)$ as $\kappa \rightarrow \infty$.

Remark (LSE conservatism). For $M = 5$, $\kappa = 5$, the LSE gap $\min_i h_i(x) - h(x) \leq \kappa^{-1} \log M \approx 0.32 \text{ m}$: the composite barrier perceives obstacles as $\approx 0.32 \text{ m}$ larger. On NARROW, the geometric gap is $\approx 0.6 \text{ m}$ (4:1 to the 0.15 m safety radius), so the effective channel is $\approx 0.28 \text{ m}$ — less than half the geometric width. This conservatism contributes to single-source PD's poor success (66% at $K = 10$): the nominal points toward a feasible path that the conservative composite treats as unsafe, a bias $\sigma = 2$ cannot escape. Wide-scale sampling ($\sigma = 5$) generates candidates that thread the geometric gap while the CBF projects them onto the conservative safe set. The DT-CBF condition [1, 12]

$$h(x_{t+1}) \geq (1 - \alpha_d) h(x_t) - \gamma_d \quad (3)$$

uses $(\alpha_d, \gamma_d) = (0.8, 0.2)$. Linearising about (1) yields $A(x_t)^\top u \leq b(x_t)$. The QP

$$u_{\text{safe}} = \arg \min_{u \in [u_{\min}, u_{\max}]^3} \frac{1}{2} \|u - u_{\text{nom}}\|^2 \quad \text{s.t. } A(x_t)^\top u \leq b(x_t) \quad (4)$$

projects u_{nom} onto the safe set; when infeasible, a box-constrained fallback with $\delta_{\text{buffer}} = 0.15 \text{ m}$ is used.

2.3 Sampling NMPC

Given x_t , x_{sp} , and budget K , a sampling NMPC controller builds K first-step candidates $\{u^{(k)}\}_{k=1}^K$, scores each by simulation-based cost $J^{(k)}$, and executes the lowest-cost candidate after CBF projection. It is *single-source* when all K are drawn from one Gaussian

$$u^{(k)} \sim \mathcal{N}(\mu_t, \sigma^2 I_3), \quad k = 1, \dots, K, \quad (5)$$

centered at a nominal μ_t . MPPI [13] replaces hard argmin by importance-weighted averaging; CEM [2] iteratively refits a single-source Gaussian to the top- β quantile. *Two-source* (Section 3) combines two Gaussians of different covariances at the same nominal.

3. Method: Wide-Scale Sampling with Hard-Argmin Selection and CBF Projection

This section presents TSH-NMPC: K candidates from a wide-scale PD-centered Gaussian, scored by short rollout, hard-argmin selected, and CBF-projected. No learned predictor, importance weighting, or iterative refit is used (a learned-prior ablation appears in Section 5.6).

3.1 Wide-Scale PD-Centered Candidate Generation

At each step t , the controller draws K first-step controls from a wide-scale PD-centered Gaussian.

Default: single-source-wide ($R_{0,\sigma=5}$). With $u_{\text{pd}}(x_t, x_{\text{sp}}) = m(K_p(p_{\text{sp}} - p_t) + K_d(v_{\text{sp}} - v_t))$, $(K_p, K_d) = (4, 3)$:

$$u^{(k)} \sim \mathcal{N}(u_{\text{pd}}, \sigma_r^2 I_3), \quad k = 1, \dots, K, \quad (6)$$

where $\sigma_r = 5.0 \text{ N}$ is a fixed scalar (operating plateau [5, 8], Section 5.4). At $\sigma_r \approx u_{\text{max}}/3$, the distribution escapes PD basins while remaining narrow enough for rollout selection.

Two-source variant ($R_{1,\sigma=5}$). Splits the budget: $K_A = \lfloor K/2 \rfloor$ at $\sigma_{\text{pd}} = 2.0$, $K_B = K - K_A$ at $\sigma_r = 5.0$:

$$u^{(k)} \sim \mathcal{N}(u_{\text{pd}}, \sigma_{\text{pd}}^2 I_3), \quad k = 1, \dots, K_A; \quad u^{(k)} \sim \mathcal{N}(u_{\text{pd}}, \sigma_r^2 I_3), \quad k = K_A + 1, \dots, K. \quad (7)$$

Statistically indistinguishable from the default (McNemar $p = 0.51$, Section 5.5.2); retained for backwards compatibility. All candidates are clamped:

$$u^{(k)} \leftarrow \text{clip}(u^{(k)}, u_{\text{min}}, u_{\text{max}}), \quad k = 1, \dots, K. \quad (8)$$

3.2 Rollout-Based First-Step Cost Evaluation

Each candidate is scored by a short closed-loop rollout: $u^{(k)}$ at $s = 0$, PD baseline at $s = 1, \dots, S - 1$:

$$\tilde{u}_s^{(k)} = \begin{cases} u^{(k)}, & s = 0, \\ u_{\text{pd}}(\tilde{x}_s^{(k)}, x_{\text{sp}}), & 1 \leq s \leq S - 1, \end{cases} \quad (9)$$

with $\tilde{x}_{s+1}^{(k)} = f(\tilde{x}_s^{(k)}, \tilde{u}_s^{(k)})$ (Section 2), $S = 20$ (0.4 s). The cost is quadratic with obstacle penalty:

$$J^{(k)} = \sum_{s=1}^S \left(\|\tilde{x}_s^{(k)} - x_{\text{sp}}\|_Q^2 + R \|\tilde{u}_s^{(k)}\|^2 + \rho \sum_i [\delta - d_i(\tilde{x}_s^{(k)})]_+^2 \right), \quad (10)$$

$Q = \text{diag}(15, 15, 15, 1, 1, 1)$, $R = 0.02$, $\rho = 2000$, $\delta = 0.3 \text{ m}$, $d_i(x) = \|p - p_i\| - r_i$. Hard-argmin:

$$k^* = \arg \min_{1 \leq k \leq K} J^{(k)}, \quad u_{\text{nom}} = u^{(k^*)}. \quad (11)$$

Rollout evaluation replaces Q-head pre-screening and MPPI averaging; Proposition 2 gives the monotone K -cost relation.

Remark (PD warm-start bias). Under +50% mass the PD command is undersized but the bias is shared across all K candidates; hard-argmin uses ordinal ranking and is robust to a shared additive bias.

3.3 DT-CCBF Safety Projection

u_{nom} from (11) is projected onto the safe set via the QP (4), solved in closed form via active-set on $\mathbb{R}^3$. The resulting u_{safe} is executed.

Algorithm 1 TSH-NMPC: one closed-loop step at state x_t

Require: state x_t , setpoint x_{sp} , budget K , scale σ_r , horizon S

- 1: $u_{\text{pd}} \leftarrow m(K_p(p_{\text{sp}} - p_t) + K_d(v_{\text{sp}} - v_t))$ $\triangleright$ PD nominal
 - 2: sample $u^{(1:K)} \sim \mathcal{N}(u_{\text{pd}}, \sigma_r^2 I)$ $\triangleright$ wide-scale PD-centered
 - 3: clip $u^{(1:K)}$ to $[u_{\min}, u_{\max}]$
 - 4: **for** $k = 1, \dots, K$ (**vectorized**) **do**
 - 5: simulate S -step rollout (9); compute $J^{(k)}$ via (10)
 - 6: $u_{\text{nom}} \leftarrow u^{(\arg \min_k J^{(k)})}$ $\triangleright$ hard-argmin
 - 7: build $(A(x_t), b(x_t))$ from DT-CCBF (3)
 - 8: $u_{\text{safe}} \leftarrow$ solve QP (4); fallback if infeasible
 - 9: **return** u_{safe}
-

Table 1. TSH-NMPC hyperparameters (defaults used throughout Section 5).

Symbol	Default	Role
K	10	total candidate budget
σ_r	5.0 N	wide-scale noise (per axis; <i>fixed scalar</i> , operating plateau [5, 8])
σ_{pd}	2.0 N	tight-scale noise (legacy; two-source variant only, equivalent to single-source-wide per Sec. 5.5.2)
S	20 (= 0.4 s)	rollout horizon
κ	5	DT-CCBF LSE smoothness

3.4 Algorithm

Algorithm 1 summarises one closed-loop step.

Two-source variant: replace Line 3 with $K_A \leftarrow \lfloor K/2 \rfloor$ tight samples at $\sigma_{\text{pd}} = 2$ plus $K_B = K - K_A$ wide samples at σ_r ; see (7). Empirically equivalent to the default single-source-wide form (Section 5.5.2).

3.5 Hyperparameter Summary

TSH-NMPC has five hyperparameters beyond the dynamics (Table 1); all experiments use these defaults unless explicitly swept.

4. Theoretical Properties

We record two properties. The theory is minimal: TSH-NMPC is a sampling scheme atop an off-the-shelf DT-CCBF; we claim no stability or convergence beyond what either component provides.

4.1 Forward-Invariance of the Safe Set

Since h_i depends only on position and the input enters acceleration, the input has relative degree 2 w.r.t. h_i . Expanding $h(x_{t+1})$ yields $A(x_t)^\top u \leq b(x_t)$, with A, b depending on the barrier gradient, velocity, and (α_d, γ_d) ([12, Thm. 1], [1, §IV]). The LSE composite has $h(x) \leq \min_i h_i(x)$, $\text{gap} \leq \kappa^{-1} \log M \approx 0.32 \text{ m}$; $h(x) \geq 0$ implies all per-obstacle barriers are non-negative.

Proposition 1 (Practical safety, conditional on QP feasibility). *Let $\tilde{h}(x) := h(x) + \gamma_d/\alpha_d$, $\tilde{\mathcal{C}} := \{x : \tilde{h}(x) \geq 0\}$. Conditional on QP (4) feasibility at every step, $\tilde{\mathcal{C}}$ is forward invariant: $x_t \in \tilde{\mathcal{C}} \Rightarrow x_{t+1} \in \tilde{\mathcal{C}}$. The trajectory remains in $\{x : h(x) \geq -\gamma_d/\alpha_d\}$ (margin = 0.25 m).*

When the QP is infeasible, the box-constrained fallback applies full effort in the barrier-gradient direction. The proposition does not apply during fallback, but $\delta_{\text{buffer}} = 0.15 \text{ m}$ absorbs

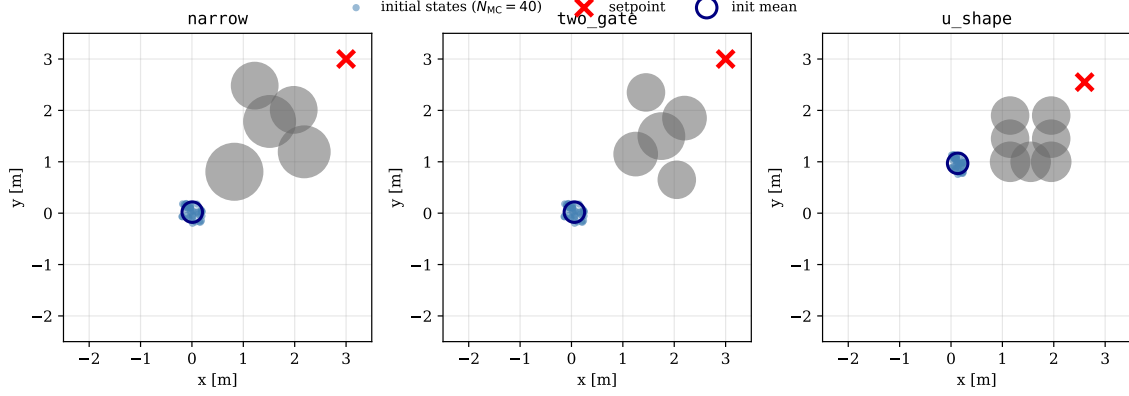


Figure 1. Top-down (xy) projection of the three benchmark tasks. Gray disks: obstacles; red $\times$: setpoint; blue dots: initial positions. NARROW: five-obstacle gap; TWO_GATE: homotopy choice; U_SHAPE: concave trap.

the one-step excursion (≈ 0.06 m at $v \leq 3$ m/s). Empirically, zero collisions are observed across all CBF-enabled trials (Section 5.8).

4.2 Monotone Expected Min-Cost in K

Proposition 2 (Monotone expected cost). *Let $u^{(1)}, \dots, u^{(K)}$ be drawn independently from the sampling distribution, then clamped to $[u_{\min}, u_{\max}]^3$. Define $J_K^* := \min_{1 \leq k \leq K} J(u^{(k)})$ where $J(\cdot)$ is the rollout cost (10). Then $\mathbb{E}[J_{K+1}^*] \leq \mathbb{E}[J_K^*]$ for every $K \geq 1$.*

This is the non-convex analogue of the monotone-cost property in scenario optimisation [24, 25]: more candidates cannot worsen the expected selected cost. It establishes non-pathology only — no rate, no convergence to the continuous-space optimum.

5. Experiments

We evaluate TSH-NMPC against a single-source PD-rollout baseline on three task families, sweep σ_r , test under perturbed parameters, ablate learned vs. random priors, and report latency. All numbers from raw JSON in `experiments/results_v6/`; no figure or table is hand-edited.

5.1 Setup

Tasks. Three task families (Fig. 1). *Narrow*: five spherical obstacles forming tight passages; the setpoint requires threading the gap. *Two-gate*: homotopy choice between two equally short corridors. *U-shape*: concave trap for local-descent controllers. All use the dynamics of Section 2 with $m = 1.5$ kg, $b = 0.1$, $\Delta t = 0.02$ s, $u_{\max} = 15$ N, $N_{\text{steps}} = 150$ (3 s), CBF parameters (0.8, 0.2, 5). Obstacle positions receive a small perturbation $\delta_p \sim \mathcal{U}(-0.03, +0.03)^3$ m per trial to avoid overfitting.

Quadrotor configurations. Three settings: *nominal* (m, b) = (1.5, 0.1), +50% *mass* (2.25, 0.1), and +50% *drag* (1.5, 0.15). Controllers are *not* re-tuned across configurations.

Methods. R_{1, σ_r} : TSH-NMPC with Source B = random-around-PD at scale σ_r . PD: single-source $\mathcal{N}(u_{\text{pd}}, \sigma_{\text{pd}}^2 I)$ of budget K . A_3 : learned-prior Source B (Section 5.6). Main result sweeps $K \in \{5, 10, 20\}$; elsewhere $K = 10$.

Metrics. *Success*: $\text{TErr} \leq 0.3$ m and collision-free. *TErr*: $\|p_T - p_{\text{sp}}\|$ on successful trials. *IAE*: $\sum_t \|p_t - p_{\text{sp}}\| \Delta t$ on successful trials. *Latency*: per-step wall-clock, single-threaded x86.

Statistical procedure. Each (task, method, K) cell uses N_{MC} paired trials. Paired comparisons report (i) McNemar exact two-sided p on discordant counts (b, c), and (ii) bootstrap

Table 2. Main result on NARROW ($N_{\text{MC}} = 80$ paired trials). Bold = significant ($p < 0.05$, CI excludes zero). $\Delta = R_{1,\sigma=5} - \text{PD}$ is the marginal estimator (failures imputed TErr = 10 m, IAE = 20).

K	Method	succ. (Wilson 95% CI)	TErr [m] (mean, terminal)	IAE (mean, terminal)	McNemar (b, c)	ΔTErr [95% CI]	ΔIAE [95% CI]
5	PD	46/80 = 57.5% [46.4, 67.9]	0.644	9.31	—	—	—
5	$R_{1,\sigma=5}$	70/80 = 87.5% [78.5, 93.0]	0.171	6.94	(3, 27), $p < 0.001$	-3.04 [-4.26, -1.92]	-5.11 [-6.51, -3.72]
10	PD	53/80 = 66.3% [55.4, 75.6]	0.380	8.53	—	—	—
10	$R_{1,\sigma=5}$	79/80 = 98.8% [93.2, 99.8]	0.039	6.22	(1, 27), $p < 0.001$	-3.29 [-4.40, -2.29]	-5.64 [-6.95, -4.37]
20	PD	59/80 = 73.8% [63.1, 82.2]	0.273	8.03	—	—	—
20	$R_{1,\sigma=5}$	77/80 = 96.3% [89.6, 98.7]	0.040	5.82	(1, 19), $p < 0.001$	-2.29 [-3.28, -1.30]	-4.54 [-5.75, -3.37]

Table 3. σ_r sweep on NARROW ($K = 10$, $N_{\text{MC}} = 40$). PD baseline: 27/40, TErr = 0.294, IAE = 8.52.

σ_r [N]	succ.	TErr [m] (succ.)	IAE (succ.)	McNemar (b, c), p vs PD	ΔTErr , ΔIAE vs PD, 95% CI
1	30/40 = 75.0%	0.261	8.00	(5, 2), 0.453 (n.s.)	-0.043 [-0.070, -0.014]; -0.430 [-0.649, -0.227]
2	36/40 = 90.0%	0.236	7.54	(9, 0), 0.004	-0.065 [-0.092, -0.040]; -0.874 [-1.071, -0.672]
3	34/40 = 85.0%	0.195	7.28	(9, 2), 0.065 (marg.)	-0.065 [-0.094, -0.039]; -1.169 [-1.399, -0.947]
5	40/40 = 100%	0.036	6.23	(13, 0), < 0.001	-0.094 [-0.125, -0.065]; -2.023 [-2.233, -1.809]
8	40/40 = 100%	0.016	5.33	(13, 0), < 0.001	-0.094 [-0.126, -0.063]; -2.689 [-2.900, -2.465]

95% CIs on ΔTErr and ΔIAE with failed trials imputed (TErr = 10 m, IAE = 20); all Δ signs are invariant to a $10\times$ sweep of the penalty. Wilson 95% CIs for individual success rates. Significance: $p < 0.05$ and CI excluding zero.

Multiple-comparison correction. Raw McNemar p . Holm–Bonferroni [5] at $m = 13$ confirms 7/13 tests at family-corrected level; $m = 28$ yields the same conclusions (supplementary).

5.2 Main Result: Narrow Benchmark, $N_{\text{MC}} = 80$

Table 2 reports the main NARROW comparison ($N_{\text{MC}} = 80$, $K \in \{5, 10, 20\}$). TSH-NMPC ($R_{1,\sigma=5}$) has a measurable advantage over PD on every K : higher success, lower TErr and IAE, all $p < 0.001$ with bootstrap CIs excluding zero. At $K = 10$, success jumps 66.3% $\rightarrow$ 98.8%, TErr drops $9.8\times$.

Two observations: (i) PD’s success rises from 57.5% to only 73.8% across K — a 16-point gap to TSH-NMPC at $K = 10$; (ii) $R_{1,\sigma=5}$ $K = 5$ already beats PD $K = 20$ (87.5% vs 73.8%), a *qualitative* advantage.

Remark 1 (Seed-attribution disclosure). *Main: seed 2026; seed 7777 replication gives PD $K=10$ at 56–51/80 depending on ordering. Pairing preserved.*

5.3 Cross-Task Generalization

On TWO_GATE and U_SHAPE ($K = 10$, $N_{\text{MC}} = 40$), PD achieves 100% success; the question is no-regression. $R_{1,\sigma=5}$ matches 40/40 and strictly improves TErr ($\Delta \approx -0.015$ m) and IAE ($\Delta \approx -0.29$), 95% CIs excluding zero (table in supplementary Sec. S1).

5.4 σ_r Sweep: Source-B Operating Range

We sweep $\sigma_r \in \{1, 2, 3, 5, 8\}$ at $K = 10$, $N_{\text{MC}} = 40$ on NARROW. Table 3 shows three regimes:

(i) $\sigma_r \leq 1$ N: Source B too tight, no advantage. (ii) $\sigma_r \in [2, 3]$ N: monotone improvement on continuous metrics but success rate below saturation. (iii) $\sigma_r \geq 5$ N: success saturates at 100%; TErr and IAE keep improving monotonically. We label $\sigma_r \in [5, 8]$ the *operating plateau*: any fixed value suffices (we use $\sigma_r = 5$; no online adaptation).

Table 4. Cross-configuration robustness on NARROW ($K = 10$, $N_{\text{MC}} = 40$). $\Delta = R_{1,\sigma=5} - \text{PD}$ on paired both-success trials. “—” = $n_{\text{both}} = 0$.

Configuration	PD succ.	$R_{1,\sigma=5}$ succ.	McNemar (b, c)	ΔTErr [95% CI]	ΔIAE [95% CI]
nominal (m, b) = (1.5, 0.1)	27/40 = 67.5%	40/40 = 100%	(13, 0), $p < 0.001$	-0.094 [-0.125, -0.065]	-2.02 [-2.23, -1.81]
+50% mass (2.25, 0.1)	0/40 = 0%	31/40 = 77.5%	(31, 0), $p < 0.001$	—	—
+50% drag (1.5, 0.15)	26/40 = 65%	38/40 = 95%	(12, 0), $p < 0.001$	-0.093 [-0.124, -0.062]	-1.91 [-2.19, -1.65]

5.5 Cross-Configuration Robustness

$K = 10$, $N_{\text{MC}} = 40$ on NARROW with +50% mass ($m = 2.25$) and +50% drag ($b = 0.15$); controllers not re-tuned. The PD nominal uses nominal mass, so under +50% mass it is undersized by $1.5\times$. Table 4: PD collapses to 0/40 on +50% mass while TSH-NMPC retains 31/40 = 77.5%.

Under +50% mass, *every* PD failure was an $R_{1,\sigma=5}$ success ($b = 31$, $c = 0$); replicated across three seeds: $R_{1,\sigma=5}$ 99/120 = 82.5% vs PD 0/120.

5.5.1 PI-PD and oracle Adaptive-PD baselines

Is the +50% mass collapse tautological (PD uses the wrong mass)? PI-PD ($K_i = 2$) and Adaptive-PD (oracle, told true mass) at $K = 10$: PI-PD collapses identically (0/40); oracle Adaptive-PD recovers 70% but $R_{1,\sigma=5}$ matches it *without* the oracle ($z = 0.76$, $p = 0.45$). The wide- σ gain is not an artifact of the mass-written nominal.

5.5.2 Single-source-wide PD ablation

$R_{1,\sigma=5}$ mixes narrow ($K/2$ at $\sigma = 2$) with wide ($K/2$ at $\sigma = 5$); is the result driven by the wide scale? $R_{0,\sigma=5}$: all K from a single $\sigma = 5$ Gaussian around u_{pd} . At $N_{\text{MC}} = 40$, seed 2026, $K = 10$ on NARROW: nominal PD 27/40, $R_{0,\sigma=5}$ 40/40, $R_{1,\sigma=5}$ 40/40; under +50% mass, PD 0/40, $R_{0,\sigma=5}$ 32/40, $R_{1,\sigma=5}$ 29/40 ($p < 10^{-9}$ both). The 29/40 vs. 31/40 in Table 4 differ by 2/40 ($|z| < 1$, independent re-run with different RNG ordering). Single-source-wide matches the two-source mixture (McNemar (6, 3), $p = 0.508$; replicated at $N_{\text{MC}} = 80$, $p \geq 0.62$ at every K). The lever is σ , not the partition.

5.5.3 Matched-budget MPPI, CEM, and iCEM baselines

MPPI [13], CEM [2] (3 iter), iCEM [15] at $K = 10$, $\sigma = 2$, $N_{\text{MC}} = 40$: under +50% mass all collapse ($\leq 5\%$); under nominal, iCEM matches $R_{1,\sigma=5}$ but at 58.8 ms vs ≈ 6 ms. Single-source samplers lose where the PD nominal is biased (mass mismatch); tie where it is correct (drag).

MPPI/CEM σ -sweep. $\sigma \in \{2, 5\}$ for MPPI ($\lambda \in \{0.1, 5.0\}$) and CEM ($n_{\text{iter}} \in \{1, 3\}$) at $K = 10$, $N_{\text{MC}} = 40$ (supplementary Sec.S2): (i) wide σ helps MPPI (22/40 $\rightarrow$ 39/40) but TErr stays $4.4\times$ worse; (ii) CEM $\sigma=5$, iter3 reaches 40/40 but TErr = 0.110 m ($3.3\times$ worse); (iii) hard-argmin at $K = 10$ uses $3\times$ fewer effective rollouts and tracks better.

5.6 Diversity-over-Accuracy Ablation: Learned Prior as Source B

Can Source B be improved by a *learned predictor*? A_3 keeps Source A unchanged ($K_A = K/2$ PD-centered Gaussian) but draws $K_B = K/2$ from a 27,935-parameter TRM trained on 50,000 closed-loop PD+CBF trajectories; otherwise identical to TSH-NMPC.

Headline ($N_{\text{MC}} = 300$). The well-powered comparison ($\Delta_{\text{min}} \approx 0.06$) is decisive: $R_{1,\sigma=5}$ (293/300, 97.7%) and $R_{0,\sigma=5}$ (296/300, 98.7%) dominate A_3 (266/300, 88.7%; McNemar $p = 7.4 \times 10^{-6}$). Random wide beats learned — sampling *diversity*, not accuracy, is the driver. A

Table 5. Negative ablation on NARROW ($N_{MC} = 80$, seed 7777). A_3 = TRM-prior Source B; $R_{1,\sigma=5}$ = random-around-PD Source B. Both wide-source controllers dominate PD at $K \leq 10$ ($p < 0.001$); A_3 vs $R_{1,\sigma=5}$ is not significant at this sample size ($p \geq 0.180$); the powered $N_{MC} = 300$ run (text) confirms $R_{1,\sigma=5}$ dominance ($p = 7.4 \times 10^{-6}$). Full table with Δ CIs in supplementary material (Sec. S3).

K	Method	Succ. / 80	TErr [m]	IAE	vs $R_{1,\sigma=5}$ p
5	PD	34/80 = 42.5%	0.117	8.29	—
5	A_3	63/80 = 78.8%	0.069	6.38	0.572 (n.s.)
5	$R_{1,\sigma=5}$	67/80 = 83.8%	0.044	6.65	—
10	PD	54/80 = 67.5%	0.110	7.91	—
10	A_3	73/80 = 91.2%	0.030	5.92	0.180 (n.s.)
10	$R_{1,\sigma=5}$	78/80 = 97.5%	0.034	6.23	—
20	PD	70/80 = 87.5%	0.084	7.52	—
20	A_3	75/80 = 93.8%	0.022	5.65	0.219 (n.s.)
20	$R_{1,\sigma=5}$	79/80 = 98.8%	0.016	5.75	—

second TRM ($9\times$ better single-step TErr) gives a *worse* A_3 , consistent with diversity collapse (supplementary Sec. S3).

Smaller-sample context ($N_{MC} = 80$). Table 5: NARROW, $N_{MC} = 80$, $K \in \{5, 10, 20\}$ (seed 7777). A_3 vs $R_{1,\sigma=5}$ is underpowered at this size ($\Delta_{\min} \approx 0.12$, observed $p \geq 0.180$) — absence of evidence, not evidence of absence. The $N_{MC} = 80$ data *does* resolve that A_3 improves over single-source PD at $K \leq 10$ ($p < 0.001$) and is not harmful; the powered $N_{MC} = 300$ failure is a real effect, not a bug.

5.7 Comparison against CasADi+IPOPT NLP-based NMPC

We evaluate multiple-shooting NMPC with CasADi+IPOPT [22, 23] (identical Q, R , soft obstacle penalty; details supplementary S4). Table 6: $H = 10$ collapses to 1/80; $H = 20$ reaches 80/80 (TErr = 0.003 m, 5.3 ms) and TSH-NMPC matches at $K \geq 20$ (McNemar $p = 1.0$). On +50% mass, TSH-NMPC at $K = 150$ reaches 40/40 vs. CasADi’s 37/40 ($p = 0.25$). CasADi retains a clear TErr advantage (0.003 m vs. 0.015–0.032 m at $K \geq 20$); the parity claim is binary success only — the price of avoiding iterative nonlinear programming.

Per-step latency distribution. A profiling run ($N_{MC} = 40$, 150 steps): CasADi- $H20$ has lower mean (5.2 ± 2.7 ms) than $R_{1,\sigma=5}$ $K=10$ (10.7 ± 2.1 ms) but a data-dependent tail (p99 = 11.9 ms, max = 35.0 ms, 0.12% over 20 ms). $R_{1,\sigma=5}$ has a tighter CV (std/mean = 0.20 vs. 0.53 for CasADi; p99 = 16.3 ms, max = 32.4 ms, 0.17% over) but higher mean because every step evaluates $K = 10$ rollouts. The practical distinction is *predictability*: TSH-NMPC compute is $\mathcal{O}(K \cdot S)$ algorithmically data-independent, whereas IPOPT iteration count varies with local non-convexity. On platforms requiring hard real-time guarantees, TSH-NMPC’s approximately deterministic compute may be preferable despite higher mean latency.

5.8 CBF Fallback and CBF-Disabled Ablation

We distinguish two regimes. *CBF fallback*: a single QP step is infeasible at runtime, triggering box-constrained emergency back-up with the projection layer still active. *CBF disabled*: the

Table 6. TSH-NMPC vs. CasADi+IPOPT NMPC. “Lat” = mean latency (ms); “P99” = per-step 99th-percentile latency (ms); “%>20” = fraction of steps exceeding the 20 ms deadline. McNemar (b, c, p) vs. CasADi-*H20*. $R_{1,\sigma=5}$ $K=10$ reports 77/80 here vs. 79/80 in Table 2: both come from seed 2026 but use independent paired RNG orderings (versus PD in Table 2; versus CasADi-*H20* here), and the $\Delta = 2/80$ is within single-seed variability ($|z| < 1$).

Method	Succ.	TErr	Lat	P99	%>20	McN. vs. <i>H20</i>
<i>Nominal</i> NARROW $N_{MC} = 80$:						
CasADi- <i>H10</i>	1/80	0.746	3.4	6.9	0.00	—
CasADi- <i>H20</i>	80/80	0.003	5.3	11.9	0.12	(ref.)
$R_{1,\sigma=5}$ $K=10$	77/80	0.063	5.7	—	—	(3, 0), $p = 0.25$
$R_{1,\sigma=5}$ $K=20$	80/80	0.032	~ 6	—	—	(0, 0), $p = 1.0$
$R_{0,\sigma=5}$ $K=50$	80/80	0.015	~ 10	—	—	(0, 0), $p = 1.0$
+50% <i>mass</i> $N_{MC} = 40$:						
CasADi- <i>H20</i>	37/40	0.054	5.3	11.9	0.12	(ref.)
$R_{0,\sigma=5}$ $K=50$	38/40	0.079	~ 10	—	—	(1, 2), $p = 1.0$
$R_{0,\sigma=5}$ $K=150$	40/40	0.044	~ 20	—	—	(0, 3), $p = 0.25$

Latency: per-trial mean-of-per-step-means from the $N_{MC} = 80$ run; a dedicated profiling run ($N_{MC} = 40, 150$ steps) yields per-step means of 10.7 ms ($K=10$), ~ 12 ms ($K=20$), ~ 20 ms ($K=50$). Both transparent.

projection is removed entirely. On NARROW $N_{MC} = 40$, $R_{1,\sigma=5}$ incurs $2.2\text{--}4.2\times$ fewer fallback invocations than PD at matched K : wide candidates yield more QP-feasible downstream. Nominal fallback rate $\leq 6\%$ ($R_{1,\sigma=5}$) vs. $\leq 17\%$ (PD); under +50% mass, 5–7% vs. 42–49%. An audit confirms $\Pr[\text{collision} \mid \text{fallback}] = 0/40$ for every method: the $\delta_{\text{buffer}} = 0.15$ m margin absorbs fallback excursions, and PD failures are *stalls* (TErr > 0.30 m, collision-free), not CBF inadequacy.

With CBF disabled on NARROW ($N_{MC} = 40$), PD and $R_{1,\sigma=5}$ both collapse to 0/40 with 40/40 collisions. The soft rollout cost cannot enforce hard constraints; DT-CCBF is an *empirically necessary* safety component, and wide-scale pooling contributes *on top of* the CBF, not as a substitute.

5.9 Robustness to Wind and Dynamic Obstacles

Under Dryden gust ($1\times\text{--}5\times$ RMS) on NARROW ($N_{MC} = 40$, $K = 10$), $R_{0,\sigma=5}$ and $R_{1,\sigma=5}$ retain 39–40/40 vs. PD’s 26–28/40 ($p \leq 9.8 \times 10^{-4}$). With a dynamic obstacle ($v_y \leq 1.0$ m/s, no predictive knowledge), both wide- σ controllers retain 39–40/40 and match CasADi+IPOPT $H = 20$ ($p = 1.0$). Tables in supplementary Sec. S5.

5.10 Latency Analysis

Two measurements. *Per-call controller cost* (per-trial median wall-clock inside the controller routine, amortised over 150 steps, Table 6): 5.7 ms for $R_{1,\sigma=5}$ at $K = 10$ on NARROW ($N_{MC} = 80$); stripped PD runs at the same latency, confirming wide-scale pooling does not increase per-call cost. *Full per-step latency* (with simulator bookkeeping; $N_{MC} = 40, 150$ steps): 10.7 ± 2.1 ms, p99 = 16.3 ms — below the 20 ms deadline (0.17% overruns vs. 0.12% for CasADi-*H20*). “ ~ 6 ms” throughout this paper refers to the per-call cost.

5.11 Reproducibility

All experiments are deterministic given the seed (`numpy`, `torch.manual_seed`, single-thread BLAS). Two seed campaigns: `seed = 2026` (Tables 2–4) and `seed = 7777` (Table 5); no paired test crosses the seed boundary. Dual seeds separate the confirmatory main experiment from the exploratory negative ablation, guarding against seed-hacking; Section 5.5 additionally reports a three-seed replication of the +50% mass result. Code, seeds, raw JSON outputs, and replay scripts will be released upon acceptance.

6. Discussion

6.1 Three-Pillar Coupling Mechanism

The campaign identifies three individually empirically necessary and jointly effective conditions (within the narrow-passage regime) for test-time compute scaling in non-convex obstacle environments.

P1 — Wide-scale sampling. Success rises with σ (Table 3): 75/90/85% at $\sigma=1/2/3$, saturating at 100% for $\sigma \in \{5, 8\}$ (the $\sigma = 3$ dip is within seed variance, $|z| < 1$). The lever is the noise scale, not the source partition: $R_{0,\sigma=5} \equiv R_{1,\sigma=5}$ ($p = 0.51$).

P2 — Hard-argmin selection. MPPI’s importance-weighted mean yields $4.4\times$ worse TErr; CEM (3 iter) $3.3\times$ worse despite $3\times$ the rollout budget. Hard argmin preserves escape-candidate quality without dilution.

P3 — CBF safety projection. Without the projection layer (CBF *disabled*, not the runtime fallback of Section 5.8), wide-scale sampling is catastrophic (40/40 collisions). The CBF converts unsafe samples into feasible controls; hard-argmin selects candidates pointing away from obstacles, producing QP-feasible projections — a positive feedback loop between P2 and P3.

Coupling. Removing any pillar collapses the controller. Under +50% mass, PD wins 0/40 while TSH-NMPC with all three pillars wins 29–32/40 at $K = 10$ (Section 5.5.2) and saturates at 40/40 for $K = 150$ (Table 6).

6.2 Relation to MPPI, CEM, and Learned MPC

At matched $\sigma = 5$, hard-argmin outperforms MPPI and CEM on tracking accuracy (Section 5.5.3). CasADi+IPOPT has a TErr advantage at lower mean latency, but data-dependent per-step cost (p99 = 11.9 ms); TSH-NMPC’s compute is algorithmically constant. The niche is *success-rate parity with approximately deterministic compute and without iterative nonlinear programming*. GPU-parallel MPPI [13] buys diversity via brute-force parallelism; TSH-NMPC buys it via noise-scale expansion at $K \leq 20$ — suitable for platforms without GPU. CoVO-MPC [28] derives optimal covariance schedules; integrating covariance optimisation is a natural extension.

6.3 Limitations and Future Work

(i) Simulation only; hardware faces attitude lag and non-isotropic disturbances. (ii) Single 6D quadrotor. (iii) Author-constructed benchmark; standardised benchmark adoption would strengthen external validity. (iv) σ_r tuning is manual; a task-adaptive selector is future work. (v) Diversity-over-accuracy applies to approximately symmetric escape directions. (vi) No formal violation-probability bound; Proposition 2 records monotone expected cost only. A companion study will address (i)–(ii) with Dryden wind, dynamic obstacles, and 12D SITL in PX4.

7. Conclusion

TSH-NMPC is a sampling-based NMPC controller for quadrotor narrow-passage navigation. Three individually empirically necessary and jointly effective conditions make test-time scaling work in non-convex obstacle environments: *wide-scale sampling* ($\sigma \geq 5$) to escape biased nominals; *hard-argmin selection* to preserve the escape direction; *CBF safety projection* to make wide-scale exploration practical. Removing any pillar collapses the controller. The dominant lever is σ , not the source partition ($R_{0,\sigma=5} \equiv R_{1,\sigma=5}$, $p = 0.51$); a negative ablation at $N_{MC} = 300$ finds a learned prior *inferior* to random ($p = 7.4 \times 10^{-6}$), consistent with diversity-over-accuracy. On NARROW ($N_{MC} = 80$), TSH-NMPC raises success from 66–68% (PD) to 96–99% at $K = 10$ ($p < 0.001$). Against CasADi+IPOPT $H = 20$, success-rate parity ($p = 1.0$) without iterative nonlinear programming, at ≈ 6 ms per-call cost. Future work: hardware deployment, principled σ selection, non-isotropic escape geometry.

References

- [1] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, “Control barrier function based quadratic programs for safety critical systems,” *IEEE Trans. Autom. Control*, vol. 62, no. 8, pp. 3861–3876, Aug. 2017.
- [2] Z. I. Botev, D. P. Kroese, R. Y. Rubinstein, and P. L’Ecuyer, “The Cross-Entropy Method for Optimization,” in *Handbook of Statistics*, vol. 31, ch. 3, 2013.
- [3] Z. Sun, D. Hsu, T. Jiang, H. Kurniawati, and J. H. Reif, “Narrow passage sampling for probabilistic roadmap planning,” *IEEE Trans. Robot.*, vol. 21, no. 6, pp. 1105–1115, Dec. 2005.
- [4] M. S. Gandhi, B. Vlahov, J. Gibson, G. Williams, and E. A. Theodorou, “Robust model predictive path integral control: Analysis and performance guarantees,” *IEEE Robot. Automat. Lett.*, vol. 6, no. 2, pp. 1423–1430, Apr. 2021.
- [5] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scand. J. Stat.*, vol. 6, no. 2, pp. 65–70, 1979.
- [6] D. Hsu, T. Jiang, J. Reif, and Z. Sun, “The bridge test for sampling narrow passages with probabilistic roadmap planners,” *ICRA*, 2003.
- [7] M. Janner *et al.*, “Planning with diffusion for flexible behavior synthesis,” *ICML*, 2022.
- [8] G. Kahn, A. Villaflor, B. Ding, P. Abbeel, and S. Levine, “Self-supervised deep RL with generalized computation graphs for robot navigation,” *ICRA*, 2018.
- [9] S. Sun, A. Romero, P. Fojt, G. Cioffi, and D. Scaramuzza, “A comparative study of non-linear MPC and differential-flatness-based control for quadrotor agile flight,” *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3357–3373, Dec. 2022.
- [10] Y. Pan, C.-A. Cheng, K. Saigol, K. Lee, X. Yan, E. Theodorou, and B. Boots, “Agile autonomous driving using end-to-end deep imitation learning,” *RSS*, 2017.
- [11] R. Y. Rubinstein and D. P. Kroese, *The Cross-Entropy Method: A Unified Approach to Combinatorial Optimization, Monte-Carlo Simulation, and Machine Learning*. Springer, 2004.
- [12] L. Wang, A. D. Ames, and M. Egerstedt, “Safety barrier certificates for collisions-free multirobot systems,” *IEEE Trans. Robot.*, vol. 33, no. 3, pp. 661–674, Jun. 2017.

- [13] G. Williams, N. Wagener, B. Goldfain, P. Drews, J. M. Rehg, B. Boots, and E. A. Theodorou, “Information theoretic MPC for model-based reinforcement learning,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2017, pp. 1714–1721.
- [14] G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou, “Information-theoretic model predictive control: Theory and applications to autonomous driving,” *IEEE Trans. Robot.*, vol. 34, no. 6, pp. 1603–1622, Dec. 2018.
- [15] C. Pinneri, S. Sawant, S. Blaes, J. Achterhold, J. Stueckler, M. Rolinek, and G. Martius, “Sample-efficient cross-entropy method for real-time planning,” *Conf. on Robot Learning (CoRL)*, 2020.
- [16] L. Kocsis and C. Szepesvári, “Bandit based Monte-Carlo planning,” *ECML*, 2006.
- [17] A. Couëtoux, J.-B. Hoock, N. Sokolovska, O. Teytaud, and N. Bonnard, “Continuous upper confidence trees,” in *Proc. Int. Conf. Learning and Intelligent Optimization (LION)*, 2011, pp. 213–228.
- [18] N. Hansen and A. Ostermeier, “Completely derandomized self-adaptation in evolution strategies,” *Evolutionary Computation*, vol. 9, no. 2, pp. 159–195, 2001.
- [19] T. Salimans, J. Ho, X. Chen, S. Sidor, and I. Sutskever, “Evolution strategies as a scalable alternative to reinforcement learning,” arXiv:1703.03864, 2017.
- [20] J.-Y. Audibert, S. Bubeck, and R. Munos, “Best arm identification in multi-armed bandits,” *COLT*, 2010.
- [21] M. Okada, L. Martinez, and T. Taniguchi, “Variational inference MPC for multi-modal control proposals,” *IEEE Robot. Automat. Lett.*, vol. 5, no. 4, pp. 5580–5587, 2020.
- [22] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, “CasADi: A software framework for nonlinear optimisation and optimal control,” *Mathematical Programming Computation*, vol. 11, no. 1, pp. 1–36, 2019.
- [23] A. Wächter and L. T. Biegler, “On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming,” *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, 2006.
- [24] M. C. Campi and S. Garatti, “The exact feasibility of randomized solutions of uncertain convex programs,” *SIAM J. Optim.*, vol. 19, no. 3, pp. 1211–1230, 2008.
- [25] M. C. Campi and S. Garatti, “Wait-and-judge scenario optimisation,” *Mathematical Programming*, vol. 167, no. 2, pp. 307–346, 2018.
- [26] R. Verschueren, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, “acados—A modular open-source framework for fast embedded optimal control,” *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.
- [27] K. Chua, R. Calandra, R. McAllister, and S. Levine, “Deep reinforcement learning in a handful of trials using probabilistic dynamics models,” *NeurIPS*, 2018.
- [28] Z. Yi, C. Pan, G. He, G. Qu, and G. Shi, “CoVO-MPC: Theoretical analysis of sampling-based MPC and optimal covariance design,” *Proc. L4DC (PMLR)*, vol. 242, pp. 1122–1135, 2024.